## TOPIC LOOKUP — READ FIRST

| | |
|---|---|
| average power density ($S_{av}$) | POWER R&T (1,1) |
| angle of reflection ($\theta_r$) | ANGLES (1,2) |
| critical angle ($\theta_c$) | ANGLES (1,2) |
| angle of refraction ($\theta_t$, Snell) | ANGLES (1,2) |
| angle of transmission ($\theta_t$, Snell) | ANGLES (1,2) |
| tangent of Brewster angle ($\tan\theta_B$) | ANGLES (1,2) |
| Brewster angle ($\theta_B$) | ANGLES (1,2) |
| polarization for reflected/transmitted waves (at $\theta_B$) | ANGLES coral note (1,2) |
| enclosing a waveguide on both sides w/ length ($f_{mnp}$) | CAVITY RES. (1,3) |
| large aperture ($d \geq$ several $\lambda$) | APERTURE ANT. (2,3) |
| circular aperture (diam. $d$) | APERTURE ANT. HPBW tbl. (2,3) |

## $\tilde{E}$ PHASOR TEMPLATES (Air: $\varepsilon_r = \mu_r = 1$)

**Wavenumber per medium** *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ \text{(rad/m)} \quad \Rightarrow \quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

**Direction table** *$u \in \{x, y, z\}$ = axis perp. to boundary*

| Inc dir | Inc | Refl | Trans |
|---|---|---|---|
| $+\hat{u}$ | $e^{-jk_1u}$ | $e^{+jk_1u}$ | $e^{-jk_2u}$ |
| $-\hat{u}$ | $e^{+jk_1u}$ | $e^{-jk_1u}$ | $e^{+jk_2u}$ |

*$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.*

*General template (incident has pol. $\hat{e}$, amplitude $a_0$):*

$$\tilde{\mathbf{E}}^i = a_0\hat{e}\,e^{\mp jk_1u}\ \text{(V/m)} \qquad \text{(incident wave, Med 1)}$$

$$\tilde{\mathbf{E}}^r = \Gamma\, a_0\hat{e}\,e^{\pm jk_1u}\ \text{(V/m)} \qquad \text{(reflected wave, Med 1)}$$

$$\tilde{\mathbf{E}}^t = \tau\, a_0\hat{e}\,e^{\mp jk_2u}\ \text{(V/m)} \qquad \text{(transmitted wave, Med 2)}$$

*(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)*

*Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.*

**$\hat{e}$ by polarization** *plug into templates above*

| Pol. | $\hat{e}$ | Note |
|---|---|---|
| Lin. $\hat{x}$ | $\hat{x}$ | $E_y = 0$ |
| Lin. $\hat{y}$ | $\hat{y}$ | $E_x = 0$ |
| Lin. at 45° | $(\hat{x}+\hat{y})/\sqrt{2}$ | in phase |
| RHC | $\hat{x} - j\hat{y}$ | $\delta = -\pi/2$ |
| LHC | $\hat{x} + j\hat{y}$ | $\delta = +\pi/2$ |
| General | $a_x\hat{x} + a_y e^{j\delta}\hat{y}$ | elliptical |

*$\Gamma, \tau$ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.*

## TABLE 8-2 — $\Gamma, \tau, R, T$ (unitless) SUMMARY

***Med. 1** ($z < 0$) incident side; **Med. 2** ($z \geq 0$) transmitted.*

| | **Normal** ($\theta_i = 0$) | **$\perp$ pol** | **$\parallel$ pol** |
|---|---|---|---|
| $\Gamma$ refl. wav | $\frac{\eta_2-\eta_1}{\eta_2+\eta_1}$ | $\frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}$ | $\frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i}$ |
| $\tau$ trans. wav | $\frac{2\eta_2}{\eta_2+\eta_1}$ | $\frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}$ | $\frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i}$ |
| $\tau$ rel. | $\tau = 1+\Gamma$ | $\tau_\perp = 1+\Gamma_\perp$ | $\tau_\parallel = (1+\Gamma_\parallel)\frac{\cos\theta_i}{\cos\theta_t}$ |
| $R$ refl. pwr | $\lvert\Gamma\rvert^2$ | $\lvert\Gamma_\perp\rvert^2$ | $\lvert\Gamma_\parallel\rvert^2$ |
| $T$ trans. pwr | $\lvert\tau\rvert^2\frac{\eta_1}{\eta_2}$ | $\lvert\tau_\perp\rvert^2\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$ | $\lvert\tau_\parallel\rvert^2\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$ |
| $R+T$ total | $T = 1-R$ | $T_\perp = 1-R_\perp$ | $T_\parallel = 1-R_\parallel$ |

*Notes: $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; nonmag. $\Rightarrow$ $\eta_2/\eta_1 = n_1/n_2$.*

**Intrinsic impedance** $\eta_i$ *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

*Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.*

**Low-loss correction** $\eta_c$ *$\sigma/(\omega\varepsilon) \ll 1$*

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

*For $\Gamma/\tau$ just use $\eta_0/\sqrt{\varepsilon_r}$. The $j$ term is for $|\eta_c|, \theta_\eta$. See LOSSY MEDIA — 5 CASES.*

## POWER REFLECTION & TRANSMISSION

**Avg power density of plane wave** *lossless; $|E_0|$ peak E-field amp.*

$$S_{av} = \frac{|E_0|^2}{2\eta}\ \text{(W/m}^2\text{)}$$

*$\eta$ = intrinsic impedance of the medium ($\Omega$):* $\eta = \sqrt{\mu/\varepsilon} = \eta_0\sqrt{\mu_r/\varepsilon_r} \xrightarrow{\mu_r=1} \eta_0/\sqrt{\varepsilon_r}$; $\eta_0 = 120\pi \approx 377\,\Omega$ *(free space).*

**% reflected** *always* $\%P_r = 100\,|\Gamma|^2$

**% transmitted** *$\eta$-ratio matters!* $\%P_t = 100\,|\tau|^2\,\eta_1/\eta_2$

*Check:* $\%P_r + \%P_t = 100$.

## ANGLES — REFLECTION, TRANSMISSION, CRITICAL, BREWSTER

**Reflection angle (Snell, reflection)** *always* $\theta_r = \theta_i$ (from normal)

**Transmission/refraction (Snell, refraction)** *single boundary; angles from normal*

$$\sin\theta_t = \sin\theta_i\sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin\theta_i\,\frac{n_1}{n_2}$$

**Multilayer chain (Snell, chained)** *stack of slabs $1 \to 2 \to 3 \to 4$*

$$\sin\theta_{i+1} = \sin\theta_i\sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

**Air–slabs–air shortcut** *outer media identical; collapse of Snell chain*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

**Critical angle** $\theta_c$ *TIR when $n_1 > n_2$; no transmitted wave*

$$\sin\theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} \Rightarrow \theta_c = \arcsin\left(\frac{n_2}{n_1}\right)$$

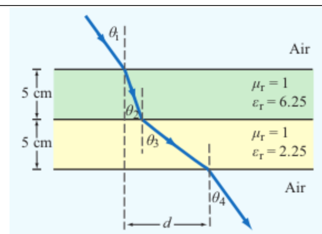
**Brewster angle** $\theta_B$ **($\parallel$-pol only)** *$\Gamma_\parallel = 0$; nonmag*

$$\tan\theta_B = \frac{n_2}{n_1} = \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} \Rightarrow \theta_B = \arctan\left(\frac{n_2}{n_1}\right)$$

*At $\theta_B$: reflected wave is pure $\perp$-pol ($\parallel$ reflection = 0); transmitted has both.*

*Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients ($\parallel$ vs $\perp$).*

## LATERAL DISPLACEMENT




**Beam offset through $N$ slabs** *thickness $t_i$, refracted angle inside slab $i$* **Slab-indexed** *$\theta_i$ = angle inside slab $i$; $t_i$ = slab thickness (m)*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_i$$

*The angle ($\theta_i$) in the formula is always the one the ray makes* inside that slab of height ($t_i$).

## LOSSY MEDIA — 5 CASES

**Loss tangent** *$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon_0 = 8.854\times10^{-12}$ F/m*

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

| Type | $\sigma/\omega\varepsilon$ | $\eta_c$, $\alpha$, $\beta$ |
|---|---|---|
| Lossless ($\sigma = 0$) | 0 | $\eta = \eta_0/\sqrt{\varepsilon_r}$ exact<br>$\alpha = 0$, $\beta = \omega\sqrt{\varepsilon_r}/c$ |
| Low-loss | $< 10^{-2}$ | $\eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)$<br>$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}$ |
| Quasi-cond. | $10^{-2}$–$10^2$ | exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$<br>see below |
| Good cond. | $> 10^2$ | $\eta_c \approx (1+j)\sqrt{\pi f\mu/\sigma}$<br>$\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}$ |
| Magnetic | any | keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$ |

*For $\Gamma/\tau$: drop $j$ correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep $\mu_r$.*

**Low-loss quick params** *$\sigma/(\omega\varepsilon) \ll 1$, $\mu_r = 1$*

$$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}\ \text{(Np/m)},\ \ \beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}\ \text{(rad/m)},\ \ \eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega)$$

**Exact (quasi-cond)** *$X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$*

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1}, \quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}$$

$$\theta_\eta = \frac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}$$

## RECTANGULAR WAVEGUIDE — MODES

**TE mode** *$E_z = 0$, $H_z \neq 0$* **TM mode** *$H_z = 0$, $E_z \neq 0$*

*Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.*

**$E_x$ form (TE)** *read $m, n$ from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$$

*coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.*

**Identify TE$_{mn}$ vs TM$_{mn}$** *cos/sin pattern of $E_x$ is identical for both!*

1. Read the problem statement (e.g. "a TE wave" $\to$ TE).
2. If $m = 0$ or $n = 0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).
3. Source field $H_z(x, y)$ given $\Rightarrow$ TE. Source field $E_z(x, y)$ given $\Rightarrow$ TM.

*TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. So TE$_{10}$ exists but TM$_{10}$ doesn't.*

## TABLE 8-3 — FULL FIELD COMPONENTS

*$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, $Z$ in $\Omega$.*

**TE mode** *generator: $\tilde{H}_z$*

$$\tilde{H}_z = H_0\cos\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TE}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TE}}$$

$$Z_{\text{TE}} = \eta/\sqrt{1 - (f_c/f)^2}$$

**TM mode** *generator: $\tilde{E}_z$*

$$\tilde{E}_z = E_0\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)E_0\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)E_0\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TM}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TM}}$$

$$Z_{\text{TM}} = \eta\sqrt{1 - (f_c/f)^2}$$

**TEM plane wave** *for reference*

$$\tilde{E}_x = E_0\,e^{-j\beta z}, \quad \tilde{H}_y = \tilde{E}_x/\eta$$

$$\eta = \sqrt{\mu/\varepsilon}\ (\Omega), \quad f_c\ \text{N/A}, \quad k = \omega\sqrt{\mu\varepsilon}$$

**Properties common to TE/TM**

$$f_c = \frac{u_{p0}}{2}\sqrt{(m/a)^2 + (n/b)^2}\ \text{(Hz)}$$

$$\beta = k\sqrt{1 - (f_c/f)^2}\ \text{(rad/m)}$$

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1-(f_c/f)^2}}\ \text{(m/s)}$$

## CUTOFF FREQUENCY

**Common cutoffs** ($a > b$) *$f_c$ formula in Table 8-3*

| Mode | $f_{mn}$ |
|---|---|
| $u_{p0}$ | $c/\sqrt{\varepsilon_r}$ (hollow: $u_{p0} = c$) |
| TE$_{10}$ | $f_{10} = u_{p0}/(2a)$ (dominant) |
| TE$_{01}$ | $f_{01} = u_{p0}/(2b)$ |
| TE$_{20}$ | $f_{20} = u_{p0}/a = 2f_{10}$ |
| TE$_{11}$,TM$_{11}$ | $f_{11} = \frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$ |

*Propagation requires $f > f_{mn}$. In $Z_{TE}/Z_{TM}$ and $u_g$, $f_c = f_{mn}$ of the excited mode.*

## CAVITY RESONATOR (guide closed both ends, length $d$)

**Resonant frequencies** *$m, n, p$ = mode indices; standing wave in all 3 dims*

$$f_{mnp} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}\ \text{(Hz)}$$

*TE$_{mnp}$: $p \geq 1$, at least one of $m, n \geq 1$. TM$_{mnp}$: $m, n \geq 1$, $p \geq 0$.*

*Dominant TE$_{101}$ when $a \geq d > b$:* $f_{101} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{1/a^2 + 1/d^2}$.

## $\varepsilon_r$ FROM DISPERSION

**Given $\omega$, $\beta$, mode $(m, n)$** *result is unitless*

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]\ \text{(unitless)}$$

*Inputs: $\omega$ (rad/s), $\beta$ (rad/m), $a, b$ (m), $c = 3\times10^8$ (m/s), $m, n$ (unitless).*

## GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0}\sqrt{1 - (f_{mn}/f)^2}\ \text{(m/s)}, \quad u_p u_g = u_{p0}^2$$

$$t = L/u_g\ \text{(s)} \quad \text{travel time through guide of length } L$$

*Higher modes have lower $u_g$ $\Rightarrow$ arrive later.*

## BOUNDARY

$\eta_1, \eta_2$ — imp. ($\Omega$)
$\Gamma, \tau, R, T$ — coeffs (unitless)
$\tau = 1+\Gamma$; $R = |\Gamma|^2$
$k_i = \omega\sqrt{\varepsilon_{r,i}}/c$ (rad/m)
$\alpha_2$ (Np/m), $\beta_2$ (rad/m): lossy
$z = 0$ — boundary plane (m)
$\eta_0 = 120\pi \approx 377\,\Omega$

## SNELL / OBLIQUE

$\theta_i, \theta_t$ — angles from normal
$n_i = \sqrt{\varepsilon_{r,i}}$ — index of refr.
$t_i$ — slab thickness (m)
$d$ — lateral disp. (m)
plane of inc. = plane of $k$+normal
$\perp$: $\mathbf{E} \perp$ this plane
$\parallel$: $\mathbf{E} \parallel$ this plane

## WAVEGUIDE

$a, b$ — wide, narrow dim. (m); $a > b$
$m, n$ — mode indices (int.)
$f_{mn}$ — cutoff (Hz)
$\beta$ — prop. const (rad/m)
$Z_{\text{TE/TM}}$ — wave imp. ($\Omega$)
$E_x$ — E-field comp. (V/m)
$H_y$ — H-field comp. (A/m)
$\eta = \eta_0/\sqrt{\varepsilon_r}$

## VELOCITIES

$u_{p0} = c/\sqrt{\varepsilon_r}$ — bulk phase vel.
$u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$
$u_g = u_{p0}\sqrt{1 - (f_c/f)^2}$
$u_p u_g = u_{p0}^2$
$t = L/u_g$ — travel time
$c = 3\times10^8$ m/s

## POWER & UNITS

$\%P_r, \%P_t$ — pwr refl./trans. (unitless)
$\%P_r = 100|\Gamma|^2$
$\%P_t = 100|\tau|^2\eta_1/\eta_2$
$\%P_r + \%P_t = 100$
$\sigma$ (S/m), $\varepsilon$ (F/m), $\mu$ (H/m)
$\varepsilon_0 = 8.85\times10^{-12}$ F/m
$\mu_0 = 4\pi\times10^{-7}$ H/m

## DIPOLE TYPE BY LENGTH
**$l$ = antenna rod length (m)**

$$\boxed{\lambda \text{ (wavelength)} = \frac{c}{f} = \frac{3\times10^8}{f} \text{ (m)}}$$

| $l/\lambda$ | Type | Use formula |
|---|---|---|
| $< 1/50$ | Hertzian (short) | HERTZIAN $S(R,\theta)$, $D=1.5$ |
| $= 1/4$ | Quarter-wave | ARB formula, $\pi l/\lambda = \pi/4$ |
| $= 1/2$ | Half-wave | half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\,\Omega$ |
| $= 1$ | Full-wave | ARB, $D=2.41$, $R_{\text{rad}}=199\,\Omega$ |
| other | Arbitrary | ARB formula |

## HERTZIAN DIPOLE — $S(R,\theta)$

**Far-field E,H phasors** *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

**Avg power density** *$l/\lambda < 1/50$; far field*

$$\boxed{S(R,\theta) = \frac{\eta_0\, k^2 I_0^2\, l^2}{32\pi^2 R^2}\sin^2\theta} \quad (\text{W/m}^2)$$

*Equivalent:* $S = S_{\max}\sin^2\theta$ *with* $S_{\max} = \eta_0 k^2 I_0^2 l^2/(32\pi^2 R^2)$ *at* $\theta = \pi/2$.

| | |
|---|---|
| $l$: rod length (m) | $I_0$: peak current (A) |
| $R$: obs. distance (m) | $\theta$: angle from dipole axis (rad) |
| $k = 2\pi/\lambda$ (rad/m) | $\eta_0 = 120\pi \approx 377\,\Omega$ |

**Total radiated power** $P_{\text{rad}} = \frac{\eta_0\pi}{3}(I_0 l/\lambda)^2$ (W)

## ARBITRARY-LENGTH DIPOLE — $S(\theta)$

**Avg power density** *any $l$; far field*

$$\boxed{S(R,\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2} \quad (\text{W/m}^2)$$

*At $\theta = 0, \pi$: bracket is $0/0$. L'Hôpital $\Rightarrow$ $S = 0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\to$ tiny $\to 0$.*

**Normalized pattern** *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

## COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

**Conversion** *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

| ° | rad | $\sin\theta$ | $\cos\theta$ | $\sin^2\theta$ | $\cos^2\theta$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 |
| 30 | $\pi/6$ | $1/2$ | $\sqrt{3}/2$ | $1/4$ | $3/4$ |
| 45 | $\pi/4$ | $\sqrt{2}/2$ | $\sqrt{2}/2$ | $1/2$ | $1/2$ |
| 60 | $\pi/3$ | $\sqrt{3}/2$ | $1/2$ | $3/4$ | $1/4$ |
| 90 | $\pi/2$ | 1 | 0 | 1 | 0 |
| 180 | $\pi$ | 0 | $-1$ | 0 | 1 |

*Antenna formulas usually take $\theta$ in rad. RAD mode on calc.*

## LINEAR ANTENNA ARRAY
**$N$ = # of elements, $d$ = interelement spacing (m)**

**Array factor (uniform, broadside)** *$k = 2\pi/\lambda$, $\gamma = kd\cos\theta$*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\max} = N^2 \text{ at } \theta = \pi/2$$

**Normalize** $F_a^{\text{norm}} = F_a/N^2$

**HPBW (broadside, uniform)** *use $Nd$, not $(N-1)d$*

$$\beta \approx 0.886\,\frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ\cdot\frac{\lambda}{Nd}$$

**Electronic steering (scan to $\theta_0$)** *replace $\gamma$ with $\gamma' = kd(\cos\theta - \cos\theta_0)$*

$$\delta = \frac{2\pi d}{\lambda}\cos\theta_0 \left(\frac{\text{rad}}{\text{elt}}\right); \quad \theta_0 = \frac{\pi}{2}\text{:broadside, 0:end-fire}$$

**Frequency scan** *feed-line incr. $\Delta\ell = n_0\lambda_0$*

$$\cos\theta_0 \approx \frac{n_0\lambda_0}{d}\cdot\frac{\Delta f}{f_0}$$

*Grating lobes appear if $d > \lambda$. Avoid.*

## BEAM PARAMETERS — $\beta$, $\Omega_p$, $D$

**Normalize** *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

**Beamwidth $\beta$ at threshold $X$**

Set $F(\theta_X) = X$, solve for $\theta_X$; $\beta = 2\theta_X$ (symmetric)

| Threshold | $X = 10^{\text{dB}/10}$ | $\theta_X$ |
|---|---|---|
| HPBW ($-3$ dB) | 0.5 | $\sqrt{\frac{\ln 2}{a}}$ |
| $-6$ dB | 0.25 | $\sqrt{\frac{\ln 4}{a}}$ |
| $-10$ dB | 0.1 | $\sqrt{\frac{\ln 10}{a}}$ |
| any $F(\theta_X)$ | any $X$ | $F(\theta_X) = X$ |

**Pattern solid angle $\Omega_p$** *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\,d\theta\,d\phi \quad \text{(sr)}$$

*No $\phi$ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\,d\theta$.*

*$\theta$ is the integration variable — DON'T plug in a number for $\theta$ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.*

| Pattern $F(\theta)$ | $\Omega_p$ (sr) |
|---|---|
| Isotropic ($F = 1$) | $4\pi \approx 12.57$ |
| Hertzian $\sin^2\theta$ | $8\pi/3 \approx 8.38$ |
| Half-wave dipole | $\approx 7.66$ |
| Narrow Gauss $e^{-a\theta^2}$ | $\pi/a = \pi\theta_{1/2}^2/\ln 2$ |
| 2-axis HPBW (aperture) | $\beta_{xz}\cdot\beta_{yz}$ |
| Other (use integral) | numerical |

*TI-36X: `[2nd][d/dx]` for $\int_\square^\square$, RAD mode, multiply by $2\pi$.*

**Directivity $D$** *always $D = 4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p} \quad \text{(unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

*WORD TRAP: "direction" = $\theta_{\max}$ (rad); "directivity" = $D$ (unitless).*

**Gain (if asked)** *$\xi$ = radiation efficiency (unitless), $0 < \xi \le 1$*

$$G = \xi D \text{ (unitless)} \qquad G_{\text{dB}} = 10\log_{10} G \text{ (dB)}$$

*Lossless / ideal antenna* $\Rightarrow G = D$.

$\xi = \frac{R_{rad}}{R_{rad} + R_{loss}} = \frac{P_{rad}}{P_{rad} + P_{loss}}$; $R_{rad} = 2P_{rad}/I_0^2$ $(\Omega)$; $R_{in} = R_{rad} + R_{loss}$.

## COMMON DIPOLE PARAMETERS

*Lossless ($\xi = 1$): $G = D$.*

| Type | $l$ | $D$ | $D_{\text{dB}}$ | $R_{\text{rad}}$ ($\Omega$) |
|---|---|---|---|---|
| Isotropic | N/A | 1 | 0 | — |
| Hertzian | $< \lambda/50$ | 1.5 | 1.76 | $80\pi^2(l/\lambda)^2$ |
| Half-wave | $\lambda/2$ | 1.64 | 2.15 | 73.2 |
| Monopole | $\lambda/4$ (gnd) | 1.64 | 2.15 | 36.6 |
| Full-wave | $\lambda$ | 2.41 | 3.82 | 199 |

*Half-wave: $P_{rad} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{rad}$, $R_{rad}$ halved.*

## ANTENNA AREAS ($A_e$, $A_p$)

**Effective area** *antenna's RX cross section; $D$ from BEAM PARAMS or table above*

$$A_e = \frac{\lambda^2 D}{4\pi} \quad (\text{m}^2)$$

**Physical cross section** *wire dipole, diameter $d$; $l$ see table above*

$$A_p = l\,d \quad (\text{m}^2\text{, any dipole})$$

**Inverse:** $D$ **from** $A_e$

$$D = \frac{4\pi A_e}{\lambda^2} \quad \text{(unitless)}$$

*Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).*

## FRIIS TRANSMISSION FORMULA

*For each antenna's $G$: if given by NAME ("half-wave dipole" etc.) $\to$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\to$ use as given (convert dB $\to$ linear).*

**Pwr density at RX, then RX pwr** *$G_t, G_r$ linear gains; $\lambda = c/f$*

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad \boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

*Equivalent:* $P_r = S_r A_{e,r}$.

**Solve for $R$** *range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

**Equivalent form using $A_e$** *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

*Free-space path loss:* $L_{fs} = (4\pi R/\lambda)^2$, *so* $P_r = G_t G_r P_t / L_{fs}$.

**General form (off-axis)** *when patterns NOT peak-aligned*

$$P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t\, F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

*$F_t, F_r$ normalized patterns at the link directions; $F = 1$ when aligned.*

**Recipe** 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.
3. Get $G_t, G_r$ by antenna type:
- named dipole $\to$ COMMON DIPOLE PARAMS ($G = D$ lossless)
- aperture / dish $\to$ APERTURE ANTENNAS ($G = D \approx 4\pi A_p/\lambda^2$)
- arbitrary pattern $F(\theta)$ $\to$ BEAM PARAMS ($G = \xi\cdot 4\pi/\Omega_p$)
- given in dB $\to$ dB$\leftrightarrow$LINEAR table

4. Plug into Friis.

*Use linear $G$, not dB. Convert FIRST.*

## dB $\leftrightarrow$ LINEAR

*Works for any power-like quantity ($G$, $D$, $P_r/P_t$, SNR, ...):*

$$X_{\text{dB}} = 10\log_{10} X_{\text{lin}} \qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

*Memorize: 3 dB $\Leftrightarrow$ $\times 2$, 10 dB $\Leftrightarrow$ $\times 10$. Add dB = multiply linear.*

*For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).*

## NOISE & SNR

**Noise power** *thermal noise into matched receiver*

$$P_N = k_B\, T_{\text{sys}}\, B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; $T_{\text{sys}}$ system noise temp (K); $B$ receiver bandwidth (Hz).

**Signal-to-noise ratio**

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

## APERTURE ANTENNAS (horn, dish; large aperture, $d \ge$ several $\lambda$)

*$A_p$ physical aperture area (m$^2$); $A_e$ effective area (m$^2$, $= \lambda^2 D/4\pi$); $\beta_{xz}, \beta_{yz}$ HPBWs in two planes (rad); $\ell, d$ aperture side / diameter (m).*

*Far-field req.: $R \ge 2d^2/\lambda$ ($d$ = largest aperture dim.).*

**Pattern (rect., uniform illum.)** *peak at $\theta = 0$ (broadside)*

$$F(\theta) = \text{sinc}^2\left(\frac{\pi\ell_x\sin\theta}{\lambda}\right); \quad S_0 = \frac{E_0^2 A_p^2}{2\eta_0\lambda^2 R^2}$$

**Directivity (any shape, uniform illum.)**

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

**Directivity from beamwidths** *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

**HPBW and $D$ by aperture shape** *uniform illum.; $\beta$ in rad*

| Shape | $A_p$ | HPBW (rad) | $D$ (unitless) |
|---|---|---|---|
| Rect. $\ell_x \times \ell_y$ | $\ell_x\ell_y$ | $\beta_x \approx 0.88\lambda/\ell_x$<br>$\beta_y \approx 0.88\lambda/\ell_y$ | $D \approx 4\pi/(\beta_x\beta_y)$ $= 4\pi\ell_x\ell_y/\lambda^2$ |
| Square $\ell$ | $\ell^2$ | $\beta \approx 0.88\lambda/\ell$ | $D \approx 4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$ |
| Circular (diam. $d$) | $\pi d^2/4$ | $\beta \approx 1.02\lambda/d$ | $D \approx 4\pi/\beta^2 = \pi^2 d^2/\lambda^2$ |

**Scaling (any ratio)** *$D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$*

$$D_{\text{new}} = D_{\text{old}}\cdot\frac{A_{p,\text{new}}}{A_{p,\text{old}}}\cdot\left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}}\cdot\sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}}\cdot\frac{f_{\text{old}}}{f_{\text{new}}}$$

*If a parameter is unchanged, its ratio $= 1$. Examples: double area $\to$ $D\times 2$, $\beta \times 1/\sqrt{2}$; double freq $\to$ $D \times 4$, $\beta \times 1/2$.*

## DIPOLE

$l$ — antenna length (m)
$\lambda = c/f$ (m); $c = 3\times10^8$ m/s
$I_0$ — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
$R$ — obs. dist. (m); $\theta$ from axis
$\eta_0 = 120\pi \approx 377\,\Omega$
$S(R,\theta)$ — pwr density (W/m$^2$)

## BEAM

$F = S/S_{\max}$ (unitless)
$a$ — coef. on $\theta^2$ in Gaussian
$\beta$ — HPBW (rad or °)
$\Omega_p$ — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
$\xi$ radiation eff. (unitless); $G = \xi D$
$D_{\text{dB}} = 10\log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

## FRIIS / dB

$P_t$ — power transmitted (W)
$P_r$ — power received (W)
$G_t$ — transmit gain (linear)
$G_r$ — receive gain (linear)
$S_r = G_t P_t/(4\pi R^2)$ (W/m$^2$)
$P_r = G_t G_r(\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB $= \times 2$, 10 dB $= \times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

## APERTURE

$A_p$ — physical area (m$^2$)
$A_e = \lambda^2 D/(4\pi)$ (m$^2$)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p/\lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

## ARRAY

$N$ — number of elements
$d$ — spacing (m)
$\gamma = kd\cos\theta$
$F_a = \sin^2(N\gamma/2)/\sin^2(\gamma/2)$
$F_a^{\max} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.886\lambda/(Nd)$ (broadside)